Dr. Utkarsh Jadli

Electronics & Industrial Instrumentation Engineer
Brisbane, QLD, Australia

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Work Authorisation: Full working rights in Australia

PROFESSIONAL SUMMARY

PhD-qualified Electronics Engineer with 4+ years of experience designing and deploying intrinsically safe (Intrinsic Safety / IS) industrial instrumentation into active mining environments. Led end-to-end development of three production-grade downhole systems including strain acquisition, borehole survey, and RPM sensing modules. Specialist in precision analog front-end design, embedded firmware development (STM32, ESP32), PCB layout in Altium, and IEC 60079 / IEC 80079 hazardous area compliance. Experienced in full product lifecycle delivery from schematic design through field validation and certification.
Portfolio: utkarshjadli-bit.github.io

CORE COMPETENCIES

Intrinsic Safety (IS) Design — IEC 60079 Compliance — Hazardous Area Electronics
Industrial Instrumentation — Downhole Systems — Embedded Systems Engineering
Precision Analog & Mixed-Signal Design — PCB Development (Altium)
Firmware Development (Embedded C) — HART — RS-485 — Modbus RTU
Mining Electronics Deployment — Field Validation — Systems Testing

PROFESSIONAL EXPERIENCE

Sigra Pty Ltd – Brisbane, Australia *Nov 2021 – Present*
Electronics & Field Engineer

Industrial Instrumentation Development & Deployment

- Led end-to-end development and deployment of three intrinsically safe mining instruments operating in hazardous underground environments.
- Designed a 24-bit strain acquisition system achieving 17.8-bit effective resolution and 1.83 microstrain RMS noise within a 27 mm cylindrical housing.
- Developed IMU-based borehole survey system achieving +/- 0.33 degree azimuth accuracy, +/- 0.11 degree inclination accuracy and +/- 0.25 degree toolface accuracy
- Validated HART telemetry over 1.4 km armoured drilling cable across three independent tool builds.
- Designed a passive Hall-effect switching sub-system that uses the core barrel's own magnetic field to gate battery power — zero manual intervention required downhole.

Intrinsic Safety & Compliance

- Designed energy-limited circuits compliant with IEC 60079 and IEC 80079 standards for hazardous area applications.
- Performed worst-case capacitance analysis, component derating, and safety documentation for certification.
- Supported IS compliance reviews and technical documentation for industrial deployment.

Systems Engineering & Test Infrastructure

- Designed closed-loop hydrostatic pressure test rig (0–22 MPa) using PID-controlled electropneumatic regulator. Measured Young's modulus of aluminium: 70.8 GPa (2.65% error).
- Conducted 67-hour thermal drift testing achieving 22.50 ppm per degree Celsius stability after reference subtraction.
- Performed mechanical and strain validation using UCS testing equipment.

Embedded Firmware & Industrial Communication

- Developed embedded firmware in Embedded C for STM32 and ESP32-S3 microcontrollers.
- Implemented HART, RS-232, RS-485, and Modbus RTU communication protocols for industrial telemetry.

- Integrated sensor subsystems including strain gauges, IMUs, thermistors, Hall-effect sensors, and time-of-flight modules.

Project Delivery & Field Support

- Delivered $250k+ AUD ACARP drilling project on time and within budget during global semiconductor shortages.
- Managed alternative component qualification and adaptive PCB redesign.
- Conducted on-site troubleshooting across active mining sites under operational time constraints.

RESEARCH & ENGINEERING EXPERIENCE

Griffith University – Brisbane, Australia *2018 – 2021*
PhD Researcher – Power Semiconductor Devices

- Characterised Si MOSFETs, SiC MOSFETs, and GaN HEMTs using I-V and C-V measurement techniques.
- Developed SPICE-based semiconductor device models for power electronics simulation.

QMNC, Griffith University – Brisbane, Australia *2019 – 2021*
Research Assistant / Fellow

- Designed and debugged PCBs in Altium for laboratory sensor systems.
- Supported interdisciplinary research teams with electronics integration and fault diagnosis.

EDUCATION

Ph.D. – Power Semiconductor Devices, Griffith University (Brisbane) 2018 – 2021
M.Tech – Control Engineering, Graphic Era University (Dehradun) 2015 – 2017
B.Tech – Electrical & Electronic Engineering, Graphic Era University (Dehradun) 2011 – 2015

TECHNICAL SKILLS

PCB Design: Altium Designer, KiCAD, Analog and Mixed-Signal Design, EMI Mitigation, DFM/DFT
Embedded Systems: STM32, ESP32, Embedded C, PID Control
Industrial Communication: HART, RS-232, RS-485, Modbus RTU, 4–20 mA
Instrumentation: 24-bit Strain Systems, IMU Calibration, Hydrostatic Testing, Industrial Sensors
Simulation & Modelling: LTspice, PSpice, QSpice, MATLAB, Simulink, Python (NumPy, SciPy)
Intrinsic Safety: IEC 60079, IEC 80079, Hazardous Area Design, Energy Limiting Circuits

PUBLICATIONS & PROFESSIONAL MEMBERSHIP

Published research in semiconductor device modelling and power electronics systems.
Full list: https://scholar.google.com.au/citations?user=kmwQvPQAAAAJhl=enoi=ao
Member: IEEE Power Electronics Society — IEEE Electron Device Society